BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 275 · Database tables, entities and keys

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Define INNER JOIN.

Q2. What is the purpose of a primary key?

Q3. Which SQL clause filters rows?

Q4. Insert 12 into BST with root 10 and right child 15. State its final position.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

entity — Type of identifiable object.

attribute — Descriptive field of an entity.

primary key — Unique row identifier.

foreign key — Reference to key in another table.

index — Search aid on selected columns.

New learning

Understand — An entity corresponds to a distinguishable object type; a relational table stores rows of its attributes.

Apply — A primary key uniquely identifies each row; foreign keys express references from related tables.

Evaluate — A secondary/index key can accelerate searching without necessarily being unique. One-to-many relationships use a foreign key on the many side.

Worked example

Identify primary and foreign keys in Customer(CustomerID, Name) and Order(OrderID, CustomerID).

1. Predict first — Orders have their own identity and store a reference to the customer who placed them.

2. Model answer — Explain the reasoning and justify the output.

We Do · Practise with guidance

A pupil may submit several Homework rows. Choose PK and FK.

```
Pupil(PupilID PK); Homework(SubmissionID PK,PupilID FK)
```

Predict first

The Homework many-side contains PupilID referencing the one-side table.

Model answer

Explain the reasoning and justify the output.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Define entity, attribute, primary key, foreign key. [4]

Question 2

For Student, Book, Loan given above, state PKs and both FKs. [6]

Question 3

Explain why storing every loan and every pupil detail in a single repeated row causes anomalies. [6]

Exit Ticket · Answer before checking

A foreign key must always be unique across every row in its table.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Combines matching table rows.

Q2. Uniquely identifies each row.

Q3. WHERE.

Q4. 12>10→right to15; 12<15→left child of15.

You Do pages · Independent questions

Q1. Entity object type [1]; attribute property [1]; PK unique row identifier [1]; FK relation to key [1].

Q2. Student.StudentID [1], Book.BookID [1], Loan.LoanID [1]; Loan.StudentID→Student [1], Loan.BookID→Book [1]; one student may have many loans [1].

Q3. Pupil data repetition [1], update inconsistency [2], insertion difficulties [1], deletion loses data [1], separate related tables with keys solve [1].

Exit Ticket · Model answer

False. Multiple loan rows can reference one student.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first.

GREEN PEN Then: tick and rewrite mistakes.